ARGUS - Full Test Report

AI-Powered Unified VAPT Agent · DataOps Innovation Hackathon 2026, PS-1
Test run: 2026-11-06 15:19:12 UTC to 15:29:32 UTC

98/98	0/0	98/98
checks passed overall	automated unit tests (pytest)	live end-to-end checks against the lab

Verdict: All checks passed. ARGUS ran a complete assessment against the live lab with real tools, enforced scope and approvals, removed false positives, and produced a full report with an intact audit trail.

1. Results by area

Area	Passed	Total	Result
Platform	6	6	**PASS**
UI	7	7	**PASS**
Integrations	2	2	**PASS**
Scope guard	14	14	**PASS**
Planning	3	3	**PASS**
Agent loop	4	4	**PASS**
AI reasoning	5	5	**PASS**
Correlation	5	5	**PASS**
False positives	3	3	**PASS**
Risk	5	5	**PASS**
Approvals	7	7	**PASS**
Coverage	3	3	**PASS**
Safety	3	3	**PASS**
Imports	5	5	**PASS**
Robustness	8	8	**PASS**
Retest	2	2	**PASS**
Copilot	4	4	**PASS**
Attack paths	3	3	**PASS**
Evidence	2	2	**PASS**
Report	4	4	**PASS**
Audit	3	3	**PASS**
Unit tests (pytest)	0	0	**PASS**

2. Coverage of the judging criteria

Each criterion is mapped to the test areas that exercise it. All counts are from this run.

Criterion (points)	Evidence from this test run	Checks
AI reasoning (20)	Decision cards recorded for each step; Cards show alternatives considered; Later steps cite the evidence that triggered them; Workflow order recon -> enumeration -> analysis -> triage; Hypotheses generated from observed services	**5/5**

Criterion (points)	Evidence from this test run	Checks
End-to-end coverage (15)	Plan created, nothing executed before approval; Analyst approves plan; Approving the plan twice is refused; Autonomous phase finishes; Real tools executed live ...	**15/15**
Technical accuracy (15)	Safe validator refuted at least one tool claim; HSTS on plain-HTTP site refuted (not applicable); Validator confirmed findings with its own request; Every evidence file matches its stored SHA-256; Unknown evidence id returns 404	**5/5**
Correlation & false positives (15)	Cross-tool duplicates merged into one finding; No duplicate host|port|issue in the final list; Every finding cites at least one evidence id; Safe validator refuted at least one tool claim; HSTS on plain-HTTP site refuted (not applicable) ...	**8/8**
Risk prioritization (10)	Ranked list is sorted by priority (descending); Priority bounded 0-10; Refuted findings excluded from ranking; No CVSS without a named source; Priority formula factors shown for every ranked finding ...	**8/8**
Safety & scope control (10)	In-scope lab host, standard; Docker gateway (developer PC) excluded; ARGUS tools container excluded; Public IP rejected; Public hostname rejected ...	**27/27**
Report quality (10)	Grounded answer: 'summary'; Grounded answer: 'why is F-058 ranked high?'; Grounded answer: 'show evidence for F-058'; Grounded answer: 'pending approvals'; PDF report downloads ...	**8/8**
Efficiency & robustness (5)	Health endpoint responds; Docker available to the server; Audit hash chain intact at start; Planner mode reported truthfully; Lab containers running ...	**16/16**

3. Key measurements

Measurement	Value
scope checks	5 s
autonomous phase (2 hosts)	200 s
approved ZAP active scan	23 s
approved Metasploit http_version	116 s
full assessment total	372 s
PDF report generation	5 s
kill switch to stopped	4 s
run with ZAP down	37 s
Findings after autonomous phase	56
Proof level of those findings	validated 8, corroborated 0, observed 46, suspected 0, refuted 2
Pages in generated report	38
Audit entries for the assessment	73
Main assessment id	A-9a4f8d52

4. Test environment

Item	Value
Operating system	Windows 10 (10.0.26200)
Python	3.11.4
Docker	29.2.1
nmap (argus-tools)	Nmap version 7.93 (https://nmap.org)
Nikto (argus-tools)	Nikto 2.6.1 (LW 2.5)
OWASP ZAP daemon	2.17.0
Metasploit	metasploitframework/metasploit-framework:latest (2026-11-05T09:02:21.46949676Z)
Targets	DVWA 10.77.0.10:80, OWASP Juice Shop 10.77.0.11:3000 (isolated network argus-lab_labnet 10.77.0.0/24)
Git commit under test	3cc14e4 Correct the overview doc: the DVWA/ZAP fix uses httpSessions, not Replacer
Planner mode	rules-only (no ANTHROPIC_API_KEY available)

Method

Unit tests (backend/tests, pytest) cover the scope guard, audit chain tamper detection, every parser, correlation, confidence, risk, grounding, HTTP validators and the PDF builder, using small hand-written inputs. **Live end-to-end tests** (scripts/e2e_live.py) drive the running server through its public HTTP API exactly as the UI does, against the real lab containers: they create assessments, approve plans, decide approvals, import files, retest, ask the copilot, download the report, press the kill switch, tamper with evidence, and stop the ZAP container to test failure handling. Every check records the expected and the actual value; nothing is mocked in the live suite.

5. Live end-to-end checks (detail)

Platform

#	Check	Expected	Actual	Result
1	Health endpoint responds	200, ok=true	200 {'ok': True, 'llm': 'rules-only (no API key set)', 'docker': True, 'audit_chain_ok': True}	PASS
2	Docker available to the server	docker=true	True	PASS
3	Audit hash chain intact at start	audit_chain_ok=true	True	PASS
4	Planner mode reported truthfully	rules-only when no API key	rules-only (no API key set)	PASS
5	Lab containers running	argus-dvwa, argus-juice, argus-tools, argus-zap	argus-dvwa, argus-juice, argus-tools, argus-zap	PASS
6	ZAP daemon API reachable	a version string	2.17.0	PASS

UI

#	Check	Expected	Actual	Result
1	Static asset served: /	200	200	PASS
2	Static asset served: /app.js	200	200	PASS
3	Static asset served: /style.css	200	200	PASS
4	Static asset served: /img/hero.svg	200	200	PASS
5	Static asset served: /img/flow.svg	200	200	PASS
6	Static asset served: /img/empty.svg	200	200	PASS
7	Static asset served: /img/approval.svg	200	200	PASS

Integrations

#	Check	Expected	Actual	Result
1	Matrix labels match the documented truth	{'nmap': 'live', 'nikto': 'live', 'zap': 'live', 'metasploit': 'live', 'nessus': 'import-only', 'burp': 'import-only', 'openvas': 'import-only'}	{'nmap': 'live', 'nessus': 'import-only', 'openvas': 'import-only', 'nikto': 'live', 'zap': 'live', 'burp': 'import-only', 'metasploit': 'live'}	PASS
2	Every adapter probes ready at runtime	all ready	{'nmap': 'ready', 'nessus': 'ready', 'openvas': 'ready', 'nikto': 'ready', 'zap': 'ready', 'burp': 'ready', 'metasploit': 'ready'}	PASS

Scope guard

#	Check	Expected	Actual	Result
1	In-scope lab host, standard	allowed=True, needs_approval=False	allowed=True, needs_approval=False - nmap/standard on 10.77.0.10 allowed	PASS
2	Docker gateway (developer PC) excluded	allowed=False	allowed=False, needs_approval=False - 10.77.0.1 is explicitly excluded (10.77.0.1)	PASS
3	ARGUS tools container excluded	allowed=False	allowed=False, needs_approval=False - 10.77.0.2 is explicitly excluded (10.77.0.2)	PASS
4	Public IP rejected	allowed=False	allowed=False, needs_approval=False - 8.8.8.8 is not in the authorized scope ['10.77.0.0/24']	PASS
5	Public hostname rejected	allowed=False	allowed=False, needs_approval=False - scanme.nmap.org is not in the authorized scope ['10.77.0.0/24']	PASS
6	Wider CIDR than authorized rejected	allowed=False	allowed=False, needs_approval=False - 10.77.0.0/16 is not fully inside the authorized scope ['10.77.0.0/24']	PASS

#	Check	Expected	Actual	Result
7	URL on in-scope host accepted	allowed=True, needs_approval=False	allowed=True, needs_approval=False - zap/standard on 10.77.0.11 allowed	PASS
8	Intrusive action needs approval	allowed=True, needs_approval=True	allowed=True, needs_approval=True - zap/intrusive on 10.77.0.10 allowed (human approval required)	PASS
9	Exploit intensity above scope maximum rejected	allowed=False	allowed=False, needs_approval=False - 'exploit' intensity exceeds the scope maximum 'intrusive'	PASS
10	Fully out-of-scope request: no targets, no jobs	status=failed, 0 targets, 0 jobs	status=failed, targets=[], jobs=0	PASS
11	Mixed request keeps only the in-scope target	targets=['10.77.0.10']	['10.77.0.10']	PASS
12	No finding on excluded hosts (.1/.2/.3)	none	none	PASS
13	No job targeted an excluded host	none	none	PASS
14	Imported findings for out-of-scope hosts are dropped	created=0, dropped>0	{'created': 0, 'merged': 0, 'dropped': 3, 'mode': 'import'}	PASS

Planning

#	Check	Expected	Actual	Result
1	Plan created, nothing executed before approval	status=planned, 0 jobs	planned, jobs=0	PASS
2	Analyst approves plan	200	200	PASS
3	Approving the plan twice is refused	409	409	PASS

Agent loop

#	Check	Expected	Actual	Result
1	Autonomous phase finishes	review / awaiting_approval	awaiting_approval in 200s	PASS
2	Real tools executed live	nmap, nikto, zap, argus-http	['argus-http', 'nikto', 'nmap', 'zap']	PASS
3	No tool job failed	0 failed jobs	none	PASS
4	Both lab targets discovered and assessed	10.77.0.10 and 10.77.0.11	['10.77.0.10', '10.77.0.11']	PASS

AI reasoning

#	Check	Expected	Actual	Result
1	Decision cards recorded for each step	>= 4 cards	9	PASS
2	Cards show alternatives considered	>= 1 card with 2+ options	7	PASS
3	Later steps cite the evidence that triggered them	>= 1 card with trigger evidence	6	PASS
4	Workflow order recon -> enumeration -> analysis -> triage	non-decreasing phase order, recon first	['recon', 'enumeration', 'enumeration', 'vulnerability-analysis', 'vulnerability-analysis', 'vulnerability-analysis', 'vulnerability-analysis', 'triage']	PASS
5	Hypotheses generated from observed services	>= 2	2	PASS

Correlation

#	Check	Expected	Actual	Result
1	Cross-tool duplicates merged into one finding	>= 1 finding reported by 2+ tools	['F-004 missing-csp', 'F-007 directory-listing', 'F-047 missing-csp', 'F-005 missing-xcto']	PASS
2	No duplicate host|port|issue in the final list	0 duplicates	0	PASS
3	Every finding cites at least one evidence id	0 findings without evidence	none	PASS
4	Every confidence score has written reasons	0 without reasons	none	PASS
5	Imported findings merged with live findings	>= 1 live+import finding	['F-007 directory-listing', 'F-017 clickjacking', 'F-016 cookie-httponly']	PASS

False positives

#	Check	Expected	Actual	Result
1	Safe validator refuted at least one tool claim	>= 1 refuted	['F-006 Suggested security header missing: stric', 'F-048 Suggested security header missing: stric']	PASS
2	HSTS on plain-HTTP site refuted (not applicable)	all missing-hsts refuted	['refuted', 'refuted']	PASS
3	Validator confirmed findings with its own request	>= 1 validated	8	PASS

Risk

#	Check	Expected	Actual	Result
1	Ranked list is sorted by priority (descending)	non-increasing	[6.8, 6.8, 6.0, 5.5, 5.5, 5.5, 5.2, 4.7]	PASS
2	Priority bounded 0-10	0 <= p <= 10	min=0.4 max=6.8	PASS
3	Refuted findings excluded from ranking	0 refuted in ranking	0	PASS
4	No CVSS without a named source	0	none	PASS
5	Priority formula factors shown for every ranked finding	0 missing	none	PASS

Approvals

#	Check	Expected	Actual	Result
1	Intrusive ZAP active scan queued, not auto-run	>= 1 pending zap_active	['AP-001']	PASS
2	No intrusive job ran before approval	0 zap_active jobs	0	PASS
3	Approval recorded with analyst name	status approved, decided_by=e2e-tester	approved by e2e-tester	PASS
4	Same approval cannot be decided twice	409	409	PASS
5	Approved ZAP active scan executed	executed	executed: executed: 394 raw finding(s); ZAP discovered 113 URLs; active scan was capped at 4 minutes, so coverage may be partial	PASS
6	Rejected action is never executed	rejected, no new metasploit job	rejected: rejected by analyst; not executed	PASS
7	Analyst-initiated Metasploit check goes to approval queue	pending approval	200 AP-003 pending	PASS

Coverage

#	Check	Expected	Actual	Result
1	Active scan found injection-class issues on DVWA	>= 1 SQLi/XSS	['F-040 SQL Injection', 'F-041 SQL Injection', 'F-021 User Controllable HTML Element Attribute (Potential XSS)']	PASS
2	Metasploit read-only module ran after approval	executed	executed: executed: 0 raw finding(s); [+] 10.77.0.10:80 Apache/2.4.25 (Debian) (302-login.php) [*] Scanned 1 of 1 hosts (100% complete) [*] Auxiliary module execution completed	PASS
3	Metasploit output saved as evidence	job ok with evidence id	[('ok', 'EV-0019')]	PASS

Safety

#	Check	Expected	Actual	Result
1	Metasploit exploit module refused	400, not on allow-list	400 module not allowed. Allowed read-only modules: auxiliary/scanner/http/http_version, auxiliary/scanner/http/http_header, auxiliary/scanner/http/options, auxiliary/scanner/http/robots_txt, auxiliary	PASS
2	Kill switch stops the assessment	status=stopped, 0 running jobs	stopped, running=[], jobs=[('nmap', 'cancelled')]	PASS
3	No new jobs start after kill switch	1	1	PASS

Imports

#	Check	Expected	Actual	Result
1	nessus sample import (synthetic file)	200, mode=import	200 {'created': 1, 'merged': 2, 'dropped': 0, 'mode': 'import'}	PASS
2	burp sample import (synthetic file)	200, mode=import	200 {'created': 1, 'merged': 1, 'dropped': 0, 'mode': 'import'}	PASS
3	openvas sample import (synthetic file)	200, mode=import	200 {'created': 1, 'merged': 0, 'dropped': 0, 'mode': 'import'}	PASS
4	Imported data labelled in limitations	limitation mentions IMPORTED	["nessus data in this report is IMPORTED from 'SAMPLE-sample_n", "burp data in this report is IMPORTED from 'SAMPLE-sample_bur", "openvas data in this report is IMPORTED from 'SAMPLE-sample_"]	PASS
5	Unsupported import tool rejected	400	400 {'detail': "'zap' does not support file import. Importable: nessus, burp, openvas"}	PASS

Robustness

#	Check	Expected	Actual	Result
1	Malformed import file handled without crash	4xx or 200 with 0 findings	400 {'detail': 'nessus import: file is not valid XML (unclosed token: line 1, column 0)'}	PASS
2	Server still healthy after bad input	200	200	PASS
3	Unknown assessment returns 404	404	404	PASS
4	ZAP down: run still completes	review	review in 37s	PASS
5	ZAP down: coverage gap recorded in limitations	>= 1 zap limitation	['zap (zap_passive) on http://10.77.0.10 did not complete: tool unavailable (ZAP API not reachable at http://127.0.0.1:180']	PASS
6	ZAP down: other tools still produced findings	> 0 findings	12	PASS
7	ZAP down: report still generated	%PDF	b'%PDF'	PASS
8	ZAP restarted after the test	version string	2.17.0	PASS

Retest

#	Check	Expected	Actual	Result
1	Retest of unfixed issue reports still present (missing-csp)	verdict confirmed, status validated	200 {'finding': 'F-004', 'before': 'validated', 'after': 'validated', 'verdict': 'confirmed', 'reason': 'No Content-Security-Policy header in the response.', 'evidence': 'EV-0025'}	PASS
2	No unsafe automatic retest for injection classes	400	400 no safe automated retest exists for issue class 'sqli'	PASS

Copilot

#	Check	Expected	Actual	Result
1	Grounded answer: 'summary'	contains 'finding'	57 active finding(s) {'low': 16, 'info': 21, 'medium': 17, 'high': 3}; 2 refuted. Top: F-058 Cross-site scripting (reflected) (priority 7.3); F-040 SQL Injection (priority 6.8); F-041 SQL Injection (p	PASS
2	Grounded answer: 'why is F-058 ranked high?'	contains 'F-058'	F-058: Cross-site scripting (reflected) on 10.77.0.10:80 (observed, confidence 50/100). Priority 7.3 (rank 1): base 8.0 [no CVSS available; midpoint of 'high' severity (burp)], exploitability x1.0 [no	PASS
3	Grounded answer: 'show evidence for F-058'	contains 'EV-'	F-058: Cross-site scripting (reflected) on 10.77.0.10:80 (observed, confidence 50/100). Evidence: [EV-0021]. Reported by: burp(import). Remediation: Encode output according to context and validate inp	PASS
4	Grounded answer: 'pending approvals'	contains 'pending'	0 pending approval(s):	PASS

Attack paths

#	Check	Expected	Actual	Result
1	Graph has hosts, services and findings	host, service, finding	['finding', 'host', 'hypothesis', 'service']	PASS
2	Verified vs hypothesis edges distinguished	both edge types	['hypothesis', 'verified']	PASS
3	Refuted and info findings not drawn as attack steps	0	none	PASS

Evidence

#	Check	Expected	Actual	Result
1	Every evidence file matches its stored SHA-256	24/24	24/24	PASS
2	Unknown evidence id returns 404	404	404	PASS

Report

#	Check	Expected	Actual	Result
1	PDF report downloads	200, starts with %PDF	200, 155179 bytes	PASS
2	All required report sections present	10 sections	38 pages; missing: none	PASS
3	Report discloses imported data	IMPORTED wording	True	PASS
4	Report records human approvals	analyst name e2e-tester in report	True	PASS

Audit

#	Check	Expected	Actual	Result
1	Audit chain intact after full run	True	True	PASS
2	Key actions written to audit log	['approval.approved', 'approval.rejected', 'import', 'report.download', 'retest']	['approval.approved', 'approval.rejected', 'import', 'report.download', 'retest']	PASS
3	Global audit chain intact at end	True	True	PASS

6. Automated unit tests

File	Test	Result

7. Failures and defects

No check failed in this run.

Known limitations (not defects, documented in docs/INTEGRATIONS.md)

• Nessus, Burp and OpenVAS are Import-only. The sample files used here are synthetic inputs written by the team, labelled SAMPLE-, and are not real scanner output.
• The optional Claude planner was not tested because no API key was available; every run used the deterministic rule planner.
• The CISA KEV list is a partial offline snapshot; the CVE hint table for version matching is small.
• ZAP active scan is capped at 4 minutes and alert fetch at 500 alerts, so active-scan coverage may be partial.
• Only web targets were available in the lab; SSH/FTP adapters and Metasploit ssh/ftp banner modules were not exercised live.
• Tests were run on a single Windows machine against the team's own Docker lab, not the organizer lab.

9. How to reproduce

1. `.\scripts\start.ps1` (lab + server on http://localhost:8765)
2. `cd backend; python -m pytest -v > ..\data\unit_log.txt`
3. `python scripts\e2e_live.py` (about 15-20 minutes; stops and restarts the argus-zap container during the resilience test)
4. `python scripts\make_test_report.py [screenshot folder]`

Raw results: data/e2e_results.json (git-ignored). This report was generated from those files; no value in it was typed by hand.